DS&A EXPERIMENT 2

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Program :-

```
#include <stdio.h>

void insert(int arr[], int n)
{
    int i;

    printf("Enter elements %d:\n", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
}

void display(int arr[], int n)
{
    int i;

    printf("Given array elements are:\n");

    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");
}
```

```
int main()
{
    int arr[100], n, key, i, found = 0;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    insert(arr, n);
    display(arr, n);

    printf("Enter the element to search: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++)
    {
        if (arr[i] == key)
        {
            found = 1;
            printf("Element found at position %d\n", i + 1);
            break;
        }
    }

    if (found == 0)
    {
        printf("Element not found.\n");
    }

    return 0;
}
```

Output :-

```
Enter the size of the array: 6
Enter elements 6:
2
5
8
1
5

0
Given array elements are:
2 5 8 1 5 0
Enter the element to search: 5
Element found at position 2


=== Code Execution Successful ===
```

Input:-

```
#include <stdio.h>

int binarySearch(int arr[], int size, int key)
{
    int low = 0;
    int mid;
    int high = size - 1;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (arr[mid] == key)
            return mid;

        else if (arr[mid] < key)
```

```
            low = mid + 1;

        else
            high = mid - 1;
    }

    return -1;
}

int main()
{
    int array[] = {3, 4, 5, 6, 2, 9};
    int size = 6;
    int key;

    printf("Enter the element: ");
    scanf("%d", &key);

    int result = binarySearch(array, size, key);

    if (result != -1)
        printf("Element found at index %d\n", result);
    else
         printf("Element not found");

    return 0;
}
```

Output:-

